WORKING JOURNAL SUMMARY

Course:	03CIT4067 Cybersecurity Project
Group Number:	Group 7
Project Title:	Topic 15: Design and Implementation of a Web Application Security Risk Detection and Remediation Demonstration System Based on OWASP Top 10
GitHub Repository:	https://github.com/mustafammk242424-cell/group-7-Cybersecurity.git

1. TOPIC SELECTION NARRATIVE

Initial Alternatives and Screening Criteria

At the inception of the project, Group 7 evaluated three potential topics from the module list: (1) *Topic 3: Flask-Based System Log Anomalous Login Detection and Alerting System*; (2) *Topic 17: Design and Implementation of Secure Identity Authentication for Web Systems Based on JWT*; and (3) *Topic 15: Design and Implementation of a Web Application Security Risk Detection and Remediation Demonstration System Based on OWASP Top 10*.

We screened these against four critical criteria: **Technical Feasibility** (alignment with team proficiency in Python, Flask, and Docker), **Data and Tools Availability** (requirement for open-source, zero-cost environments), **Time Feasibility** (fit for a two-semester timeline), and **Educational Significance** (potential for tangible, code-level demonstrations). Topic 3 was deemed too narrow in scope, while Topic 17 lacked a comprehensive vulnerability demonstration environment.

Advisor Discussion and Final Decision

During our initial consultation with our supervisor, Dr. LIAM XU, we discussed the limitations of existing Dynamic Application Security Testing (DAST) tools. Dr. XU advised us to focus heavily on the "Remediation" and "Demonstration" aspects of the title, rather than just detection.

Consequently, we selected **Topic 15**. This topic survived the screening process because it allowed us to design a **Hybrid Architecture**: utilizing the intentionally vulnerable OWASP Juice Shop for broad vulnerability coverage, while developing custom Python Flask modules to specifically demonstrate the remediation of the newly emphasized OWASP 2025 category, A10 (Mishandling of Exceptional Conditions). This approach directly addresses the industry gap between automated risk flagging and secure, "fail-closed" architectural implementation.

2. MEETING DISCUSSIONS AND FOLLOW-UP ACTIONS (CLOSED-LOOP)

Meeting 1: Inception, Architecture Design, and Threat Modeling (Week 3/5)

- **Discussion:** The team finalized the project scope and proposed a Hybrid Architecture. We initiated a comprehensive STRIDE Threat Modeling analysis to map potential attack vectors (Spoofing, Tampering, Elevation of Privilege) to the OWASP Top 10: 2025 framework.
- **Action:** Developed the initial Work Breakdown Structure (WBS) and Two-Term Gantt Chart. Set up the local Docker environment for OWASP Juice Shop (for members with unrestricted network access).
- **Outcome:** Completed the Phase 1 Project Proposal draft, defining the SMART objectives and technology stack justification.
- **Verification (Next Meeting):** Reviewed the finalized System Architecture Diagram and STRIDE matrix in Meeting 2 to ensure all high-risk threats had corresponding remediation strategies planned for Phase 2.

Meeting 2: Phase 2 Implementation & Agile Risk Mitigation (Week 7 | Oct 20, 2026)

- **Discussion:** Reviewed Sprint 2 deliverables. Mustafa successfully demonstrated code-level remediation for A05 (Security Misconfiguration) and A10 (Fail-Open vs. Fail-Closed logic). *Critical Blocker Identified:* Team member Zoutao reported hospitalization and severe network restrictions in Mainland China, preventing local Docker image pulls and environment deployment.
- **Action:** As Team Leader, I initiated an Agile Risk Mitigation strategy (documented in Appendix C). We reallocated local container execution tasks to Mustafa (operating in HK). Zoutao was reassigned to focus on documentation, specifically compiling detailed Red Team operation logs (A01-A10) which did not require local Docker execution.
- **Outcome:** Successfully integrated 14 screenshots of vulnerability exploits and remediation code snippets into the Phase 2 Report. Zoutao's Red Team logs were seamlessly integrated into Chapter 5 and Appendix B.
- **Verification (Next Meeting):** Confirmed in Meeting 3 that task reallocation did not delay the project timeline and that all Chapter 5 deliverables met academic tone requirements.

Meeting 3: Phase 3 Integration, Security Testing & Proposal Finalization (Week 9 | Nov 3, 2026)

- **Discussion:** Reviewed OWASP ZAP Automated Baseline Scan results (identifying 2 Medium-risk CSP and CORS alerts, confirmed as expected default behaviors of the vulnerable Juice Shop environment). Reviewed Mustafa's Chapter 8 draft and Zoutao's academic translation of the Attack Methodology. Discussed the strict "Zero-Error / One-Shot" Blackboard submission policy.
- **Action:** Mapped technical outcomes (ZAP metrics and A10 Fail-Closed logic) into the 9 required sections of the Assignment 03 Proposal. Assigned specific presentation roles to ensure 100% member participation (Lai Man: Architecture; Mustafa: Tech Demo; Zoutao: STRIDE/Conclusion).
- **Outcome:** Finalized `AS_7_Proposal.pdf` and `AS_7_Slides.pdf` ahead of internal deadline (Nov 1, 23:59). Conducted a strict APA 7th citation check across all integrated chapters.
- **Verification (Next Meeting):** Conducted a full 15-20 minute presentation rehearsal to verify time management and Q&A readiness.

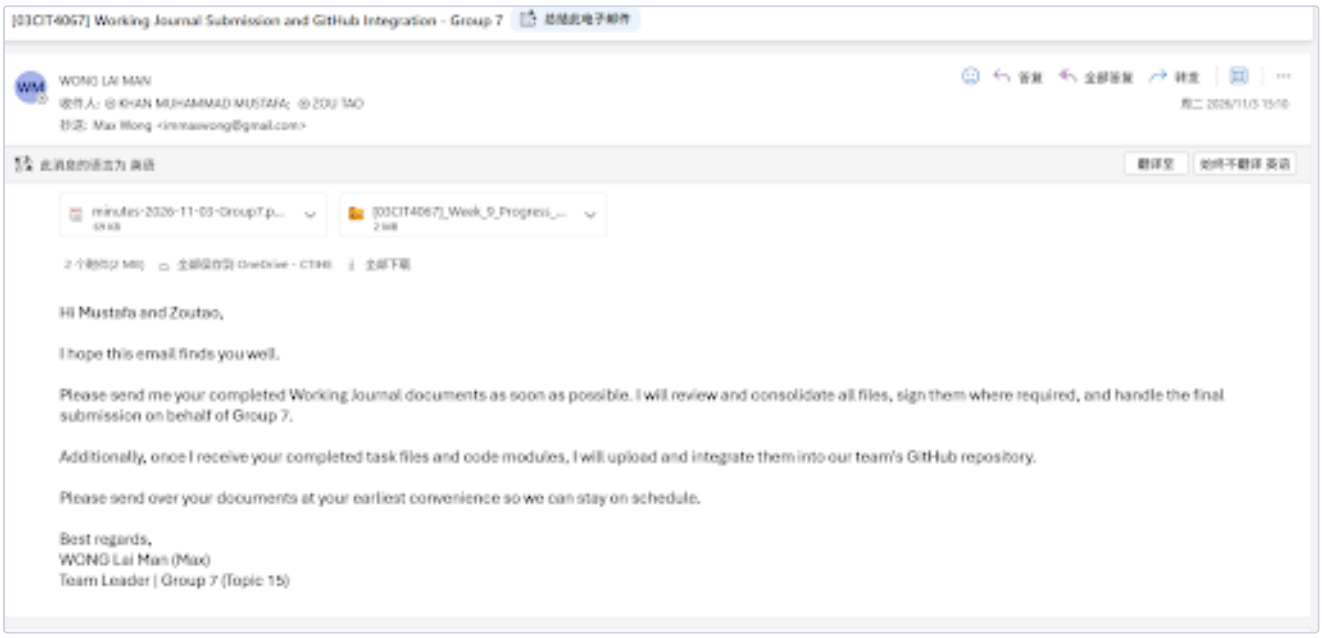
Meeting 4: Presentation Rehearsal & Q&A Strategy (Week 11)

- **Discussion:** Conducted a live run-through of the presentation deck. Identified that initial slides contained too much text ("text walls"), violating the presentation rubric. Anticipated potential Q&A questions from Dr. XU regarding how we handle ZAP "False Positives" and the specific behavior of the Fail-Closed mechanism during a total database outage.
- **Action:** Redesigned slide deck to prioritize visual aids (Architecture Diagrams, Gantt Charts, and Side-by-Side Code Comparisons). Drafted a Q&A response script focusing on our manual verification process for ZAP alerts and HTTP 503 return logic in Flask middleware.
- **Outcome:** Presentation time optimized to 18 minutes. All members demonstrated high confidence in their respective sections.
- **Verification (Next Meeting):** Final confirmation of all Blackboard submission packages and individual reflection alignment.

Meeting 5: Project Closure, Archiving, and Reflection (Week 13)

- **Discussion:** Verified that all group deliverables (Proposal, Slides, Gen AI Form, Working Journal ZIP) were successfully prepared for Blackboard submission. Discussed requirements for Individual Reflection (D4) and Peer Evaluation.
- **Action:** Each member independently drafted their Individual Reflection, ensuring coverage of the four required points (Problem solved, Advantages, Next steps, Overall thoughts) without cross-copying. Finalized GitHub repository structure (`/agenda`, `/minutes`, `/docs`, `/src`) for long-term archiving.
- **Outcome:** Successfully closed the project lifecycle. All agile artifacts, meeting minutes, and codebases were securely archived.
- **Verification:** Final submission receipts and GitHub commit histories were compiled as D3 (Timely-submission evidence) for the Working Journal.

3. TIMELY-SUBMISSION EVIDENCE (D3)


[03CIT4067] Working Journal Submission and GitHub Integration - Group 7

WONG LAI MAN

Hi Mustafa and Zoutao,

I hope this email finds you well.

Please send me your completed Working Journal documents as soon as possible. I will review and consolidate all files, sign them where required, and handle the final submission on behalf of Group 7.

Additionally, once I receive your completed task files and code modules, I will upload and integrate them into our team's GitHub repository.

Please send over your documents at your earliest convenience so we can stay on schedule.

Best regards,
WONG Lai Man (Max)
Team Leader | Group 7 (Topic 15)

Figure 1: *Email dispatch record confirming the delivery of Working Journal documents and Phase 2 deliverables by Team Leader WONG Lai Man to group members on November 3, 2026.*

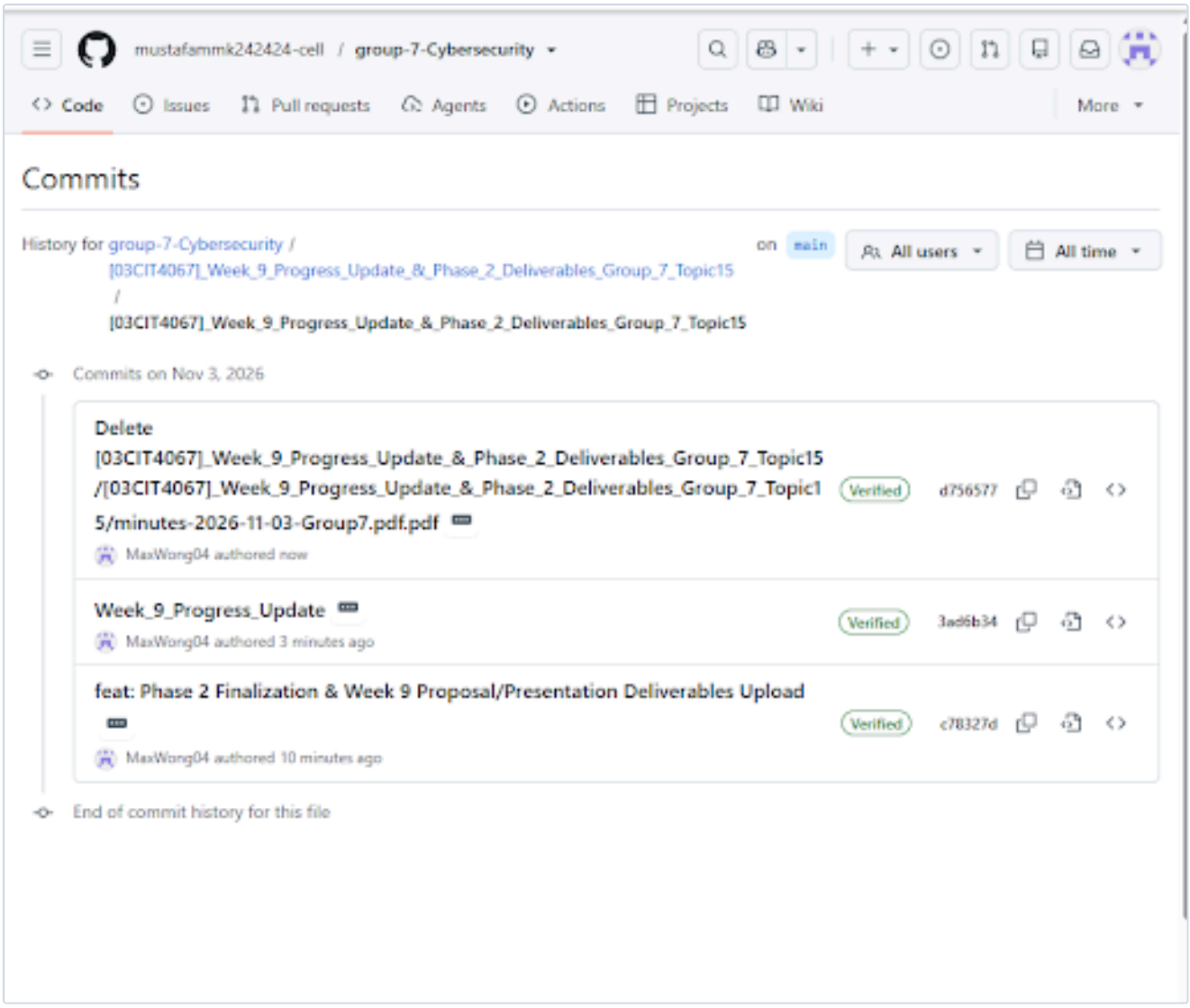


Figure 2: *GitHub repository commit history demonstrating timely version control commits for Week 9 progress updates and Phase 2 deliverables on November 3, 2026.*